

RDXXB SERIES RADAR

Modular JSON Output Schemas

Categorized by Data Type · Mix & Match · Pick What You Need

Schema Version	1.0
Release Date	21 April 2026
Total Categories	8 (incl. Basic)
Total Modules	7

0. Basic Usage Schema (Quick Start)

For most applications, you only need location, target classification, and timestamps. This Basic Usage schema combines the essential fields into one compact JSON structure. Perfect for real-time tracking, display systems, and bandwidth-constrained networks.

Includes: Radar GPS + Orientation | Target ID + Classification + Signals + Position + Velocity | UTC Timestamp

Size: ~200 bytes per target | Bandwidth: ~2 KB per target per frame | Update: Per pulse group

```
{
  "radar_location": {
    "gps": {
      "longitude": 107.608234,
      "latitude": -6.897456,
      "elevation_m": 150.0
    },
    "orientation": {
      "roll_angle_deg": 0.5,
      "pitch_angle_deg": 1.2,
      "north_correction_angle_deg": 8.5
    }
  },
  "targets": [
    {
      "track_id": 1042,
      "track_delete_flag": "0x00",
      "track_status": "active",
      "track_type_code": "0x20",
      "track_type_label": "drone",
      "confidence": 0.87,
      "energy": 2847.43,
      "credit_ratio": 0.92,
      "distance_m": 1243.7,
      "azimuth_deg": 142.3,
      "elevation_pitch_deg": 7.2,
      "height_m": 155.8,
      "radial_speed_ms": -3.4,
      "longitude": 107.612483,
      "latitude": -6.903124,
      "altitude_m": 156.0,
      "vx_east_ms": -2.1,
      "vy_north_ms": -2.7,
      "vz_up_ms": 0.3,
      "unix_timestamp": 1766049291.82
    },
    {
      "track_id": 1051,
      "track_delete_flag": "0x00",
      "track_status": "active",
      "track_type_code": "0x20",
      "track_type_label": "drone",
      "confidence": 0.74,
      "energy": 1921.05,
      "credit_ratio": 0.78,
      "distance_m": 2781.2,
      "azimuth_deg": 138.7,
      "elevation_pitch_deg": 3.9,
      "height_m": 189.4,
      "radial_speed_ms": 5.1,
      "longitude": 107.635817,
      "latitude": -6.921089,
      "altitude_m": 190.0,
      "vx_east_ms": 4.2,
      "vy_north_ms": 2.8,
      "vz_up_ms": -0.6,
      "unix_timestamp": 1766049291.82
    }
  ]
}
```

```

    }
  ]
}

```

0.1 Basic Usage Field Reference

Quick reference for all fields in the Basic Usage schema. Each field maps back to the protocol.

Field	Type	Source	Units	Purpose
radar_location.gps.longitude	float	Status frame	deg WGS-84	Radar installation longitude (6 decimals)
radar_location.gps.latitude	float	Status frame	deg WGS-84	Radar installation latitude (6 decimals)
radar_location.gps.elevation_m	float	Status frame	meters	Radar antenna elevation above sea level
radar_location.orientation.roll_angle_deg	float	Status frame	degrees	Roll angle (inclinometer X-axis)
radar_location.orientation.pitch_angle_deg	float	Status frame	degrees	Pitch angle (inclinometer Y-axis)
radar_location.orientation.north_correction_angle_deg	float	Status frame	degrees [0,360]	North correction calculated by radar
targets[].track_id	integer	Per-target field	ID	Persistent track identifier
targets[].track_delete_flag	hex	Per-target field	flag	0x00=active, 0x01=deleted
targets[].track_status	string	Derived	–	Human-readable: 'active' or 'deleted'
targets[].track_type_code	hex	Per-target: track_type	–	0x20=drone, 0x11=person, 0x12=car, etc.
targets[].track_type_label	string	Derived	–	Human-readable class: drone, person, car, bird, vessel, other, false
targets[].confidence	float	Per-target: confidence	ratio [0,1]	Autonomous ID confidence (precision 0.01). 0.87 = 87%
targets[].energy	float	Per-target: energy	dB/linear	Received signal strength
targets[].credit_ratio	float	Per-target: credit_ratio	ratio [0,1]	Track quality score
targets[].distance_m	float	Per-target: distance	meters	Slant range from radar to target
targets[].azimuth_deg	float	Per-target: azimuth	degrees [0,360]	Compass bearing to target
targets[].elevation_pitch_deg	float	Per-target: pitch	degrees	Elevation angle (negative=below horizon)
targets[].height_m	float	Per-target: height	meters	Height above reference datum
targets[].radial_speed_ms	float	Per-target: speed	m/s	Doppler velocity along bore-sight (negative=approaching)
targets[].longitude	float	Per-target: longitude	deg WGS-84	Target longitude (6 decimals)
targets[].latitude	float	Per-target: latitude	deg WGS-84	Target latitude (6 decimals)
targets[].altitude_m	float	Per-target: altitude_gps	meters WGS-84	Target GPS altitude
targets[].vx_east_ms	float	Per-target: vx	m/s	Velocity East (ENU frame)
targets[].vy_north_ms	float	Per-target: vy	m/s	Velocity North (ENU frame)
targets[].vz_up_ms	float	Per-target: vz	m/s	Velocity Up (ENU frame, positive=ascending)

targets[].unix_timestamp	float	Per-target: UTC fields	seconds	Unix timestamp (seconds since epoch) for easy application correlation
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0.2 When to Use Basic Usage

- ✓ Real-time tracking displays (map, scope, dashboard)
- ✓ Live threat detection and alerting systems
- ✓ Bandwidth-constrained networks (satellite, cellular)
- ✓ High-frequency updates (20+ fps with many targets)
- ✓ Mobile apps or IoT devices with limited storage
- ✓ Cloud transmission where bandwidth cost is high

■ **Recommendation:** *For full system diagnostics, health monitoring, or research archival, use the full modular schemas in Sections 2+ below. Basic Usage is intentionally minimal.*

1. Modular JSON Schema Overview

Beyond Basic Usage, the RDXXB radar outputs 75+ fields across multiple data categories. This section presents the complete data organized into 7 independent JSON modules. You can transmit, store, or process each module independently based on your application needs.

Data Categories (Advanced)

1. Frame Metadata — Protocol wrapper & sync info
2. Radar Location & Status — GPS, orientation, operational state
3. Radar Configuration — Filters, frequency, silent zones, auto-ID
4. Radar Health — Power, thermals, system status, firmware
5. Scan Header — Current sweep parameters
6. Track Data — Per-target measurements (classification, position, velocity)
7. Track Metadata — Age, history, confidence tracking

Use Cases (Advanced)

Use each module independently:

- Minimal tracking-only: Frame + Scan + Tracks
- Health monitoring: Frame + Location + Health
- Full telemetry: All modules combined
- Archive low-bandwidth: Frame + Location + Scan + Tracks (skip Health/Config)

2. Module Schemas with Examples (Advanced)

2.1 Module: Frame Metadata

Protocol wrapper and frame synchronization.

Use when: Always. Required for every transmission.

Bandwidth: ~40 bytes.

```
{
  "frame_metadata": {
    "frame_header": "0x55AA",
    "frame_id": "0xFF02",
    "frame_type": "search_target_information",
    "frame_count": 84231,
    "frame_timestamp_utc": "2025-12-17T08:34:51.820Z",
    "protocol_id": 1,
    "protocol_device": "RD01",
    "protocol_major_version": 2,
    "protocol_minor_version": 14,
    "checksum_valid": true
  }
}
```

2.2 Module: Radar Location & Status

GPS position, orientation, and operational state of the radar device.

Use when: Always tracking. Essential for georeferencing targets.

Bandwidth: ~200 bytes.

```
{
  "radar_device": {
    "device_id": 1,
    "device_label": "Front-End Radar 1",
    "model_name": "RD01",
    "network": {
      "ip_address": "192.168.1.3",
      "port": 7000
    },
  },
  "gps": {
    "longitude": 107.608234,
    "latitude": -6.897456,
    "elevation_m": 150.0,
    "heading_angle_deg": 45.0,
    "heading_valid": true,
    "satellite_count": 12,
    "installed_location_name": "Bandung, West Java"
  },
  "orientation": {
    "roll_angle_deg": 0.5,
    "pitch_angle_deg": 1.2,
    "north_correction_angle_deg": 8.5
  },
  "operational_state": {
    "work_mode": "0x11",
    "work_mode_label": "search",
    "fault_type": "0x00",
    "fault_label": "none",
    "servo_mode": "0x33",
    "servo_mode_label": "circular_scan",
  }
}
```

```

        "scan_mode": "0x01",
        "scan_mode_label": "mechanical_scanning"
    },
    "servo": {
        "current_azimuth_deg": 142.3,
        "current_pitch_deg": 5.1,
        "current_scan_cycle": 47
    },
    "frequency": {
        "current_code": 10,
        "current_x_band_ghz": 9.5,
        "current_ku_band_ghz": 16.2
    }
}
}

```

2.3 Module: Radar Configuration

Detection filters, silent zones, and autonomous identification settings.

Use when: System setup, periodic audit, or configuration logging.

Bandwidth: ~150 bytes.

```

{
  "radar_configuration": {
    "silent_zones": [
      {
        "zone_number": 1,
        "start_angle_deg": 180.0,
        "end_angle_deg": 200.0,
        "enabled": true
      },
      {
        "zone_number": 2,
        "start_angle_deg": 0.0,
        "end_angle_deg": 0.0,
        "enabled": false
      }
    ],
    "filters": {
      "height_min_near_zone_m": 50,
      "height_min_far_zone_m": 100,
      "height_max_m": 600,
      "speed_min_ms": 0.0,
      "speed_max_ms": 60.0,
      "range_min_m": 200,
      "range_max_m": 7500,
      "altitude_range_km": 7.5
    },
    "autonomous_identification": {
      "enabled": true,
      "max_targets": 2
    }
  }
}

```

2.4 Module: Radar Health

Power consumption, temperatures, system status, firmware versions.

Use when: Health monitoring, diagnostics, predictive maintenance.

Bandwidth: ~300 bytes.

```
{
  "radar_health": {
    "thermals": {
      "subarray_temperatures": [
        { "index": 1, "temperature_c": 45 },
        { "index": 2, "temperature_c": 44 },
        { "index": 3, "temperature_c": 46 },
        { "index": 4, "temperature_c": 45 },
        { "index": 5, "wave_controller_version": "1.2" }
      ],
      "signal_board_temperature_c": 52,
      "frequency_synthesizer_temperature_c": 48.5
    },
    "power": {
      "subarray_currents": [
        { "index": 1, "current_a": 5.4 },
        { "index": 2, "current_a": 5.3 },
        { "index": 3, "current_a": 5.5 },
        { "index": 4, "current_a": 5.4 }
      ],
      "signal_voltages": [
        { "channel": 1, "voltage_v": 12 },
        { "channel": 2, "voltage_v": 5 },
        { "channel": 3, "voltage_v": 3.3 }
      ]
    },
    "system_health": {
      "fan_status": {
        "fan1": "normal",
        "fan2": "normal",
        "fan3": "normal",
        "fan4": "normal"
      },
      "memory_status": {
        "qdr_initialized": true,
        "ddr_0_initialized": true,
        "ddr_1_initialized": true,
        "pcie_link_ok": true
      }
    },
    "firmware": {
      "fpga_version": "v3.21",
      "data_processing_version": "2.14.05.2025"
    }
  }
}
```

2.5 Module: Scan Header

Current sweep parameters and pulse group metadata.

Use when: Always with target data. Provides timing and scan context.

Bandwidth: ~30 bytes.

```

{
  "scan_header": {
    "search_azimuth_deg": 142.3,
    "search_elevation_deg": 5.1,
    "scan_cycle_count": 47,
    "pulse_group_id": 113842,
    "target_count": 2
  }
}

```

2.6 Module: Track Data (Per Target)

Classification, position, velocity, and signal quality for each detected target.

Use when: Always transmit with tracking systems. One record per target.

Bandwidth: ~280 bytes per target.

```

{
  "track": {
    "track_identification": {
      "track_id": 1042,
      "track_delete_flag": "0x00",
      "track_status": "active"
    },
    "classification": {
      "track_type_code": "0x20",
      "track_type_label": "drone",
      "confidence": 0.87,
      "confidence_pct": "87%"
    },
    "signal_quality": {
      "energy": 2847.43,
      "credit_ratio": 0.92,
      "credit_ratio_pct": "92%"
    },
    "polar_coordinates": {
      "distance_m": 1243.7,
      "azimuth_deg": 142.3,
      "elevation_pitch_deg": 7.2,
      "height_m": 155.8,
      "radial_speed_ms": -3.4,
      "radial_speed_label": "approaching"
    },
    "geodetic_coordinates": {
      "longitude": 107.612483,
      "latitude": -6.903124,
      "altitude_m": 156.0,
      "distance_from_radar_m": 1243.7,
      "bearing_from_radar_deg": 142.3,
      "altitude_above_radar_m": 6.0
    },
    "velocity_enu": {
      "vx_east_ms": -2.1,
      "vy_north_ms": -2.7,
      "vz_up_ms": 0.3,
      "total_speed_ms": 3.42,
      "horizontal_speed_ms": 3.41,
      "vertical_direction": "ascending"
    }
  }
}

```


2.7 Module: Track Metadata (Per Target)

Track age, history, confidence, and loss indicators.

Use when: Track management, filtering, prediction algorithms.

Bandwidth: ~50 bytes per target.

```
{
  "track_metadata": {
    "track_id": 1042,
    "loss_count_cpi": 0,
    "loss_reason_code": 0,
    "update_time_cpi": 113842,
    "age_pulses": 0,
    "radar_frontend_id": 1,
    "track_history": {
      "track_point_count": 38,
      "envelope_point_count": 6,
      "track_age_pulses": 38
    },
    "utc_timestamp": {
      "year": 2025,
      "month": 12,
      "day": 17,
      "hour": 8,
      "minute": 34,
      "second": 51,
      "millisecond": 820,
      "iso8601": "2025-12-17T08:34:51.820Z",
      "unix_timestamp": 1766049291.82
    }
  }
}
```

3. Message Composition Examples (Advanced)

Combine modules to create messages tailored to your use case. Examples below show different transmission profiles.

3.1 Minimal Tracking (20 KB/s for 50 targets)

Include: Frame Metadata + Radar Location + Scan Header + Track Data + Track Metadata

Exclude: Health, Configuration

Best for: High-frequency tracking streams, bandwidth-constrained networks, real-time tracking displays.

```
{
  "frame_metadata": { ... },
  "radar_device": {
    "device_id": 1,
    "gps": { ... },
    "operational_state": { ... }
  },
  "scan_header": { ... },
  "targets": [
    {
      "track_id": 1042,
      "track": { ... },
      "track_metadata": { ... }
    },
    { ... more targets ... }
  ]
}
```

3.2 Full Telemetry (50 KB/s for 50 targets)

Include: All 7 modules

Best for: Archive, diagnostics, post-hoc analysis, full system state capture.

```
{
  "frame_metadata": { ... },
  "radar_device": { ... },
  "radar_configuration": { ... },
  "radar_health": { ... },
  "scan_header": { ... },
  "targets": [
    {
      "track_id": 1042,
      "track": { ... },
      "track_metadata": { ... }
    },
    { ... more targets ... }
  ]
}
```

3.3 Health Monitoring Only (5 KB/s)

Include: Frame Metadata + Radar Location + Radar Health

Best for: Background health telemetry, alerting on failures, temperature monitoring.

```
{
  "frame_metadata": { ... },
  "radar_device": { ... },
```

```
  "radar_health": { ... }  
}
```

4. Module Reference & Bandwidth Summary (Advanced)

Module Name	Fields	Typical Size	Update Rate	Required?	Use Cases
Frame Metadata	10	40 B	Per frame	Always	Sync, framing, validation
Location & Status	18	200 B	~200 ms	Yes (tracking)	Georeferencing, state
Configuration	13	150 B	On change	No (periodic)	Audit, setup logging
Health	20	300 B	~200 ms	No (optional)	Maintenance, alerts
Scan Header	5	30 B	Per frame	With targets	Context, timing
Track Data	22	280 B	Per frame	Yes (tracking)	Position, velocity, class
Track Metadata	13	50 B	Per frame	Yes (tracking)	Age, history, confidence